

Chemical Space and Scoring Analysis of GR Ligand Generators

The custom score was constructed by normalizing each component relative to the reference dataset using MinMaxScaler. Predicted pIC_{50} values were scaled to the 0–1 range, synthetic accessibility scores were normalized and inverted so that synthetically easier molecules received higher values, QED was used directly since it already lies between 0 and 1, and the steroid-likeness penalty was inverted to reward non-steroidal scaffolds. These normalized components were combined using a weighted sum: 0.25 for QED, 0.25 for SA, 0.30 for pIC_{50} , and 0.20 for the penalty. This weighting scheme emphasizes predicted biological activity while maintaining drug-likeness and synthetic feasibility.

Across the individual components, GB_GA produced the most drug-like molecules (mean QED = 0.53), followed by the reference set (0.50) and DrugEx (0.42). Synthetic accessibility followed a similar trend: the reference set was easiest to synthesize (mean SA = 3.47), GB_GA was moderately accessible (3.88), and DrugEx molecules were slightly more complex (4.00). Both generative models produced less steroid-like molecules than the reference set, with penalties around 0.20 compared to 0.24 for ChEMBL2034. Predicted GR activity was highest for the reference ligands (mean pIC_{50} = 6.97), followed by DrugEx (6.75) and GB_GA (6.55). When these components were combined into the composite score, GB_GA achieved the highest mean value (0.78), followed by DrugEx (0.75) and the reference set (0.74), indicating that GB_GA generated the most balanced molecules overall.

Chemical-space visualizations supported these trends. In descriptor-based PCA and t-SNE maps, reference molecules formed a dense central cluster. DrugEx molecules tended to occupy regions close to this cluster, indicating conservative optimization around known GR ligand chemistry. GB_GA molecules extended into peripheral zones associated with lower SA and lower steroid-likeness, demonstrating exploration of novel, non-steroidal chemotypes. Bottom-ranked molecules from both generators appeared at the edges of chemical space, corresponding to extreme descriptor values such as high molecular weight, high polarity, or excessive flexibility.

Threshold analysis was performed using literature-based criteria ($\text{QED} \geq 0.4$, $\text{SA} \leq 4$, $\text{penalty} \leq 0.3$, and predicted $\text{pIC}_{50} \geq 6$). Code-based evaluation of the top-50 molecules from each generator showed that both DrugEx and GB_GA produced uniformly drug-like and synthetically feasible compounds, with no violations of the QED or SA thresholds. However, DrugEx had five molecules exceeding the steroid-likeness penalty and thirteen molecules with predicted pIC_{50} below 6, indicating weaker GR activity among some top-ranked candidates. In contrast, GB_GA had eight molecules above the penalty threshold but only a single molecule with predicted pIC_{50} below 6, demonstrating that GB_GA's top-ranked molecules more consistently satisfy potency criteria while remaining drug-like and synthetically accessible.

Overall, integrating the scoring analysis with chemical-space visualization provides a clear picture of generative model behavior. DrugEx tends to produce potent, reference-like GR scaffolds, while GB_GA generates more novel, non-steroidal, and synthetically accessible molecules with excellent drug-likeness. The custom score effectively captures the balance between potency, drug-likeness, synthetic feasibility, and structural novelty, and the chemical-space mapping confirms that top-ranked molecules occupy desirable regions. Taken together, GB_GA mut r 0.5 demonstrates the strongest balance of novelty and quality, whereas DrugEx GT ϵ 0.6 excels in producing molecules closely aligned with known GR ligand chemistry.

